EXPERIMENT -3

HALF ADDER AND HALF SUBSTRACTOR-BASIC GATES

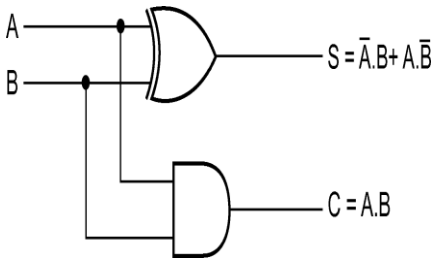
Aim 1: - To construct a Half adder circuit using basic gates

EQUIPMENTS / COMPONENTS REQUIRED :- Digital IC Trainer Kit, Ics – 7486, 7408

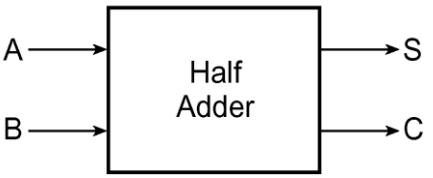
Theory:-
A half-adder is an arithmetic circuit block that can be used to add two bits. Such a circuit thus has two inputs that represent the two bits to be added and two outputs, with one producing the SUM output and the other producing the CARRY.
The Boolean expressions for the SUM and CARRY outputs are given by the equations

SUM S = A B + A B
CARRY C = A B

Procedure



A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



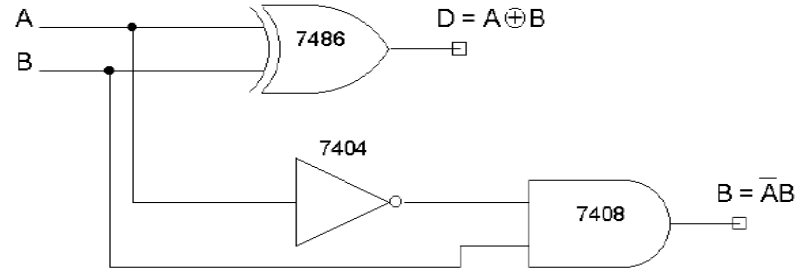
Aim 2: - To construct a Half subtracor circuit using basic gates

EQUIPMENTS / COMPONENTS REQUIRED :- Digital IC Trainer Kit, Ics – 7486, 7408,7404

Theory:-

The half subtractor is constructed using X-OR and AND Gate. The half subtractor has two input and two outputs. The outputs are difference and borrow. The difference can be applied using X-OR Gate, borrow output can be implemented using an AND Gate and an inverter.

Using X – OR and Basic Gates (a)Half Subtractor



INPUT		OUTPUT	
A	**B**	**BO**	**DIFF**
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0

RESULT: - Study and verified truth-tables of Half adder and half substractor.